

Hemodialysis: Dialysis Prescription and Adequacy

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From an idealistic perspective, an adequately treated hemodialysis (HD) patient is physically active, well nourished, and euvolemic with a maintained good quality of life and a life expectancy comparable to that of similarly aged persons with normal kidney function. The HD treatment remains a critical aspect of the multifactorial management necessary to achieve these goals. Adequacy of HD treatment today encompasses more than just the fractional replacement of excretory kidney function (dialysis dose) and extends to the prevention of intradialytic and interdialytic complications, particularly the maintenance of hemodynamic stability and symptom control. Elderly patients, the largest group of prevalent and incident dialysis patients worldwide, are especially vulnerable to hemodynamic and osmotic effects of dialysis treatment and therefore require a special approach in dialysis prescription.

ASSESSMENT OF DIALYSIS DOSE

Similar to native kidney function, the delivered dose of dialysis is conventionally assessed by the mass removal of selected uremic solutes from the patient's body. For decades, small solute clearance, assessed solely by urea as a surrogate marker, has been emphasized in clinical practice guidelines and used as measure for quality assessment and/or payment. However, evidence for relationships between small solute clearance, plasma levels of these solutes, and clinical outcomes and symptomatology is weak. Small solute clearance reflects only one of many aspects of dialysis care that are likely to affect outcomes.

Markers of Uremic Toxin Removal by Kidney Replacement Therapy

Urea Removal

Among all potential uremic toxins, only urea, a 60-Da water-soluble compound, is established as a marker of uremic solute retention and removal. Urea, which itself causes little toxicity, is a metabolite of amino acid metabolism, and urea generation therefore depends on protein intake and the balance between protein anabolism and catabolism. Urea removal was originally considered to be a good proxy for removal of pathogenic water-soluble solutes. However, it is now known that urea removal does not closely parallel the removal of other small water-soluble compounds and correlates poorly with removal of middle molecules and protein-bound solutes.¹ Nevertheless, detailed knowledge of urea kinetics is essential for the general understanding of the physics and basic principles of solute accumulation and removal in dialysis.

In a state of homeostasis, urea is evenly distributed throughout whole body water. Urea easily transfers across cell membranes by diffusion and specific urea transporters, allowing for rapid equilibration within its distribution volume after dialytic removal from the blood compartment. For almost all other water-soluble small and middle

molecular weight toxins, the intercompartmental transfer rate is much slower, leading to protracted equilibration between the various compartments and thus different removal kinetics during dialysis.² Urea removal from the body during dialysis is expressed either by urea reduction ratio (URR) or by the treatment index Kt/V . Although urea removal alone is not a sensitive marker of dialysis adequacy, it is still the standard measure to quantify removal of low molecular weight solutes during dialysis treatment. The dialysis dose ideally should be evaluated in conjunction with residual kidney function (RKF) assessed with the same uremic toxin.

Intradialytic Urea Kinetics

Dialyzers are highly efficient at removing urea from the blood, reducing plasma urea concentration during one dialyzer passage by over 90%. However, the urea clearance of the dialyzer itself is only a marker of dialyzer efficiency but not of dialysis dose, which is based on the fractional removal of urea from the whole body. The kinetics of plasma urea concentration during dialysis is satisfactorily described by a two-compartment model in which the intravascular blood compartment is replenished with urea through redistribution from extravascular compartments. The dynamics of urea redistribution depend on the urea gradient between the two compartments and on blood flow in the various tissue regions. Because intercompartmental redistribution is not immediate, urea concentration in the blood compartment during dialysis is lower than in the extravascular compartment. After a dialysis treatment, urea rebound occurs, which takes 30 to 60 minutes until full equilibration between all compartments is achieved.

Mathematical models based on the difference in predialysis and postdialysis serum urea concentration have been applied to describe intradialytic urea kinetics including URR, single-pool Kt/V (sp Kt/V), equilibrated Kt/V (e Kt/V), and weekly standard Kt/V (std Kt/V). The classic single-pool urea kinetic model (sp Kt/V) assumes single-compartment characteristics where full equilibration between blood and tissue compartments occurs immediately and the change in serum urea follows first-order kinetics, with a linear decline and no urea rebound (see Fig. 99.1, *dashed line*).³ The more physiologic "equilibrated" double-pool urea kinetic model (e Kt/V) considers the urea redistribution kinetics between the two compartments (Fig. 99.1, *solid line*).

Urea Reduction Ratio

URR, which refers to the treatment-related reduction of serum urea concentration, is a simple but rather imprecise way to quantify dialysis dose because it does not consider intradialytic urea generation and convective urea removal by ultrafiltration, and it is a single-pool measurement (Table 99.1). Despite these limitations, URR correlates quite well with dialysis outcome and is an accepted method for assessment of dialysis dose. A minimum URR of 65% to 70% is considered an adequate dialysis dose.²

Single-Pool Kt/V and Equilibrated Kt/V

The treatment index Kt/V is the most widely used parameter to assess dialysis dose. Kt/V is a dimensionless number representing the total plasma volume cleared ($K \times t$, in liters) in relation to the individual volume of distribution (V , in liters). The concept of Kt/V may be applied to any substance but in clinical practice is almost exclusively used for urea, where K is the dialyzer blood water urea clearance (in liters/hour), t is dialysis session length (hours), and V is the distribution volume of urea (in liters), which equates closely to total body water. A delivered Kt/V of 1.0 implies that the volume of plasma cleared of urea ($K \times t$) during a dialysis session is equal to urea distribution volume (V). However, because of the multicompartment kinetics of urea equilibration, and constantly ongoing urea generation, clearing a plasma volume equivalent to 1 unit of total body water (V) does not mean that all urea has been removed from body water.

In daily clinical practice, single-pool Kt/V (spKt/V) is computed from the Daugirdas equation, which is based on URR but also accounts for intradialytic urea generation and ultrafiltration volume.³ The Daugirdas equation is validated for an spKt/V range between 0.8 and

2.0 and is widely used because of its simplicity and accuracy. However, spKt/V somewhat overestimates the delivered dialysis dose because nonequilibrated postdialysis urea concentration is used for computation. The magnitude of the difference between equilibrated and nonequilibrated postdialysis urea concentration depends on the intensity of dialysis; the shorter and more intense a treatment, the higher is the postdialysis urea rebound and the larger the difference. Using eKt/V formulas, which include an estimate of equilibrated postdialysis urea concentration, grants a more accurate estimate of delivered dialysis dose. Because equilibrated post-HD serum urea concentrations are higher than nonequilibrated concentrations, eKt/V is always lower than spKt/V for any treatment (Table 99.1).

Correct assessment of Kt/V requires accurate timing of blood collections. Predialysis blood samples are collected right at the start of the treatment and postdialysis samples immediately before termination of treatment after slowing the blood flow rate to reduce any potential effect of recirculation. Blood sampling procedures should be standardized.² URR, spKt/V, or eKt/V should be assessed regularly and the dialysis prescription adjusted accordingly. In large observational studies, mortality was higher at spKt/V of less than 1.2, and a target spKt/V of 1.4 is recommended as the minimum dose for conventional thrice-weekly dialysis schedules.² Many HD machines today provide some option to monitor spKt/V online during each single dialysis session without the need for pre- or postdialysis blood sampling. However, a weakness of online methods remains the inability to measure the volume of urea distribution directly.

Weekly Dialysis Dose and Weekly Standard Kt/V

A useful way to compare native kidney function with dialysis dose across a wide variety of dialysis modalities and frequencies is by looking at an average clearance (in milliliters per minute) of a given solute achieved over a unifying period, such as 1 week. The established parameter developed for this comparison is the weekly standard Kt/V for urea (stdKt/V). With this concept, a thrice-weekly *intermittently* provided dialysis urea clearance (e.g., 250 mL/min for 4 hours thrice weekly and 0 mL/min for the remaining 6.5 days per week) is converted to an equivalent *continuous* extracorporeal urea clearance (in mL/min) related to urea distribution volume V (in milliliters).⁴ The resulting dimensionless stdKt/V value is directly compatible with the continuous native residual kidney clearance expressed as kidney Kt/V and peritoneal clearance (weekly peritoneal Kt/V). The principle of stdKt/V is visualized in Fig. 99.2, in which on the x-axis the eKt/V of a representative dialysis session is plotted onto a curve representing the prescribed treatment frequency, and the resulting stdKt/V is then derived from the intercept with the y-axis. The frequency curves are not linear because the decreasing transmembrane urea gradient over time results in decline of the efficiency in dialysis urea mass removal. Independent of the treatment schedule, a minimum stdKt/V of 2.1

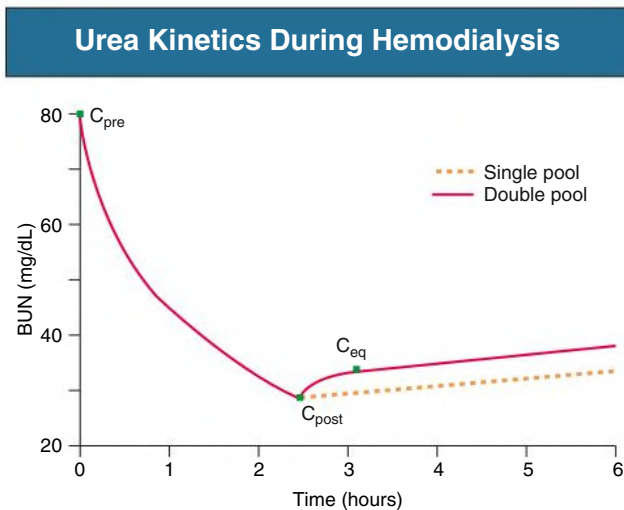


Fig. 99.1 Urea Kinetics During Hemodialysis. Urea kinetics is determined by the difference between predialysis (C_{pre}) and postdialysis (C_{post}) blood urea nitrogen (BUN). The single-pool Kt/V model assumes no relevant change in BUN concentration after termination of dialysis except for a slow constant increase as a result of BUN generation (orange dashed line). The double-pool Kt/V model (red line) more accurately describes the in vivo situation, in which intercompartmental urea redistribution leads to a urea rebound until full equilibration between blood and tissue compartments is reached. Equilibrated postdialysis BUN concentration (C_{eq}) can be mathematically predicted from postdialysis BUN (C_{post}).

TABLE 99.1 Computation of Dialysis Dose^a Results Derived From Different Model Equations

Formula	Result	Comment
$URR = (1 - C_t/C_0) \times 100\%$	67%	Urea rebound, urea generation, and ultrafiltration not taken into account
$K_t/V = \ln(C_0/C_t)$	1.10	Urea rebound, urea generation, and ultrafiltration not taken into account
$spKt/V = -\ln(R - 0.008 \times t) + (4 - 3.5R) \times UF/W$	1.33	Single-pool model; urea rebound not taken into account
$eKt/V = spKt/V - 0.6 \times spKt/V/t + 0.03$	1.16	Double-pool model for arteriovenous access, including urea rebound
$eKt/V = spKt/V - 0.47 \times spKt/V/t + 0.02$	1.20	Double-pool model for central venous access, including urea rebound

^aCalculations based on $t = 4$ hours.

$C_0 = 90$ mg/dL, $C_t = 30$ mg/dL, $UF = 3$ L, $W = 72$ kg, $R = C_t/C_0$.

BUN, Blood urea nitrogen; C_0 , predialysis BUN; C_t , nonequilibrated postdialysis BUN; eKt/V , equilibrated Kt/V; Kt/V , treatment index; $spKt/V$, single-pool Kt/V; t , dialysis duration; UF , ultrafiltration volume; URR , urea reduction ratio; W , postdialysis weight.

The stdKt/V Concept

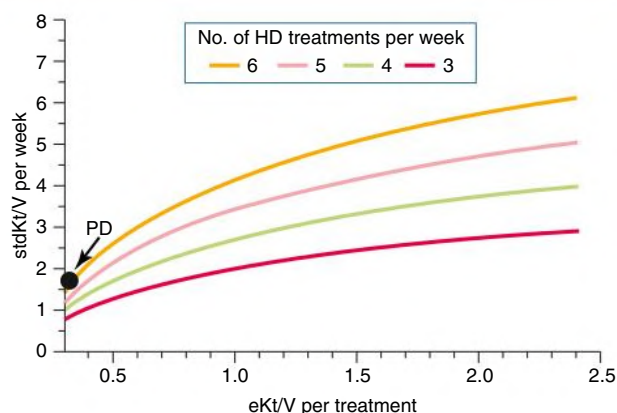


Fig. 99.2 Weekly stdKt/V Concept. The weekly stdKt/V for any hemodialysis (HD) frequency is derived by plotting eKt/V of a representative dialysis session onto the respective frequency curve. PD, Peritoneal dialysis.

accounting for kidney Kt/V is recommended for all patients.^{2,5} In anuric patients this is achieved with a thrice-weekly delivered spKt/V of 1.2 per treatment.

MIDDLE MOLECULE REMOVAL

It is widely held that retention solutes of middle molecular size may play an important role in the pathogenesis of uremia and contribute significantly to the high mortality of dialysis patients. Because of higher membrane porosity, high-flux dialyzers can remove larger amounts of middle molecules than low-flux dialyzers, and this is further increased using convective dialysis strategies, such as hemodiafiltration (HDF) and expanded HD (HDx).⁶ Because of slow intercompartmental redistribution rates, middle molecule removal is limited during conventional 4-hour dialysis sessions, and β_2 -microglobulin, a surrogate for uremic middle molecules, is more effectively removed by high-flux dialysis. Predialysis β_2 -microglobulin levels were found to be related to mortality in patients randomly assigned to high-flux or low-flux dialyzers,⁷ and certain vulnerable patient groups (e.g., those with diabetes) may gain some extra benefit from high-flux dialysis, although this is controversial.^{8,9} The European Best Practice Guidelines recommend the use of synthetic high-flux membranes to reduce cardiovascular risk and improve control of hyperphosphatemia and anemia.¹⁰

High-volume online HDF provides the largest removal of the widest range of low and middle molecular size solutes. Diffusive and convective solute elimination are maximized to the benefit of the removal of larger molecular size solutes. Because diffusive urea clearance during standard HD treatment is already high and increases by less than 10% when convective clearance is added, urea removal is not an adequate marker of treatment dose in convective dialysis modes. In HDF, mass removal of middle molecules is directly related to ultrafiltration volume, which is also referred to as *convection volume* (CV) (see Chapter 98). The *effective* CV is used as key quantifier for HDF dosing, where the term *effective* principally relates to the undiluted blood water volume removed by filtration. Thus, in postdilution mode, the effective CV is identical to total CV, whereas in predilution or mixed-dilution mode, the effective CV is considerably lower than total CV.¹¹

Various randomized controlled studies failed to demonstrate an overall survival advantage with postdilution high-volume online

Intradialytic Kinetics of Phosphate Removal and Mobilization

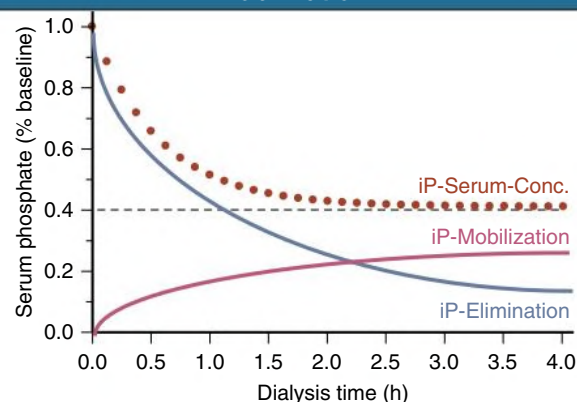


Fig. 99.3 Intradialytic kinetics of phosphate removal and mobilization. iP, Inorganic phosphate.

HDF to conventional low-flux or high-flux HD. However, an individual patient data analysis of these trials demonstrated a significantly reduced mortality risk for patients receiving more than 23 L of effective CV normalized to a body surface area of 1.73 m² in postdilution HDF mode.¹² Adjustment of target HDF dose to some measure of body size makes sense because the generation rate and distribution volume of uremic toxins are both related to body mass.

Phosphate Removal

Because hyperphosphatemia is a strong risk factor for cardiovascular mortality, predialysis serum phosphate levels should be lowered toward the normal range in all patients (see Chapter 88).¹³ Management of hyperphosphatemia is based on phosphate removal by dialysis, dietary phosphate restriction, and intestinal phosphate binding with use of phosphate binder medication. Despite its low molecular weight, the effective molecular size of phosphate (because of a rather stable coating with several water moieties) is much bigger than that of other low molecular weight solutes, which impairs intercompartmental and transmembrane phosphate redistribution during dialysis. Intradialytic phosphate kinetics resembles much more those of middle molecules, with serum phosphate levels steeply falling during the first 90 to 120 minutes into dialysis, driven by removal primarily from plasma and extracellular volumes and stabilizing thereafter (Fig. 99.3). This latter intradialytic plateau is explained by phosphate mobilization from extravascular compartments at a rate similar to that of dialyzer phosphate removal.¹⁴ Phosphate removal is improved by high-flux HD and HDF, by the use of larger dialyzer surface area, and, most dramatically, by longer or higher frequency dialysis schedules, such as short daily or daily nocturnal HD. Long, frequent dialysis schedules, such as daily nocturnal HD, offer the highest phosphate mass removal and may even result in hypophosphatemia so that phosphate needs to be added to the dialysate.

PRESCRIPTION OF HEMODIALYSIS

Hemodialysis Dose

The National Kidney Foundation's KDOQI guidelines recommend a target spKt/V of 1.4 per HD session for patients treated thrice weekly, with a minimum delivered spKt/V of 1.2. In patients with significant residual native kidney function, a lower dose of HD may be appropriate (termed *incremental dialysis*), provided RKF is measured periodically to avoid inadequate dialysis. For HD schedules other than thrice weekly, a target stdKt/V of 2.1 per week including residual kidney function is recommended.²

Delivered dialysis dose depends on dialyzer efficiency (K_d), effective treatment time (t), and distribution volume (V). Dialyzer clearance K_d depends on the flow rates within the blood and dialysate compartments (Q_b and Q_d), dialyzer K_oA , effective membrane surface area, hematocrit, and anticoagulation. Effective treatment time is essential for reaching the Kt/V target and can be substantially shorter than prescribed treatment time because of intermittent pump stops or patient demand. Urea distribution volume V does not substantially change during a single HD session but may change over time. Therefore, dialysis dose needs to be adjusted when lean body mass and thus V increases. On the contrary, if there is a loss in muscle mass, which is associated with a decrease in V , dialysis dose should not be reduced but rather based on a higher (ideal) patient V to support gains in body mass.²

A standard dialysis prescription should consist of a high-flux dialyzer, a minimum treatment time of 4 hours, a blood flow rate of at least 250 mL/min, and a dialysate flow rate of 500 to 800 mL/min. The prescription is then adjusted to meet the target $spKt/V$ of 1.4 volumes. In severe and long-standing uremia, the target dose is approached slowly over the course of several sessions to avoid the dialysis disequilibrium syndrome. Confronted with an inadequate URR or Kt/V , it is sensible to check whether the studied session was representative of the average session, because unusual problems may have occurred (e.g., shortened time, vascular access recirculation, single-needle HD). Frequent causes of inadequately low delivered dialysis dose with recirculation are vascular access problems, particularly of arterial supply. Blood sampling errors also should be considered because delayed post-HD blood sampling results in falsely low Kt/V results. If a low Kt/V remains unexplained, treatment time should be increased and a more efficient dialyzer and higher blood and dialysate flow rates should be considered (Fig. 99.4). Delivered Kt/V should be checked whenever the dialysis prescription has been modified substantially.

Hemodiafiltration Dose

The two major aims governing the prescription of high-volume online-HDF dose are adequate delivery of the target effective CV and the prevention of excess hemoconcentration. High-flux membranes offering high hydraulic permeability, as well as high middle-molecule clearance, are required for effective HDF. Hydraulic permeability is reflected by the membrane ultrafiltration coefficient, whereas membrane permeability is defined by the sieving coefficients for selected middle molecules. For high-volume online HDF a high-flux membrane with an ultrafiltration coefficient greater than 20 mL/h/mm Hg/m² and a sieving coefficient for β_2 -microglobulin of 0.6 is required.¹⁵ The major determinants of achieving a prescribed target CV in postdilution HDF are blood flow rate (Q_b), ultrafiltration rate (UFR), and treatment time (t). Because ultrafiltration is applied to undiluted blood, hemoconcentration necessarily occurs together with a lowering of blood flow rate within the dialyzer. Under certain circumstances, this leads to the deposition of plasma proteins on the membrane surface, clogging of membrane pores, and occlusion of dialyzer capillaries. To prevent excessive hemoconcentration, UFR and Q_b need to be individually adjusted so that the filtration fraction (FF), which is defined as the ratio of UFR to Q_b , does not exceed 25%. Higher FF up to 30% can be safely achieved only with modern HD machines designed to optimize filtration rate based on constant assessment and optimization of transmembrane pressure. Effective treatment time in high-volume HDF with a fixed target CV will thus always depend on achievable Q_b and UFR.

Treatment Time and Frequency

The European best practice guidelines recommend that dialysis should be delivered at least 3 times per week and the total duration should be at least 12 h/wk, unless supported by significant residual kidney

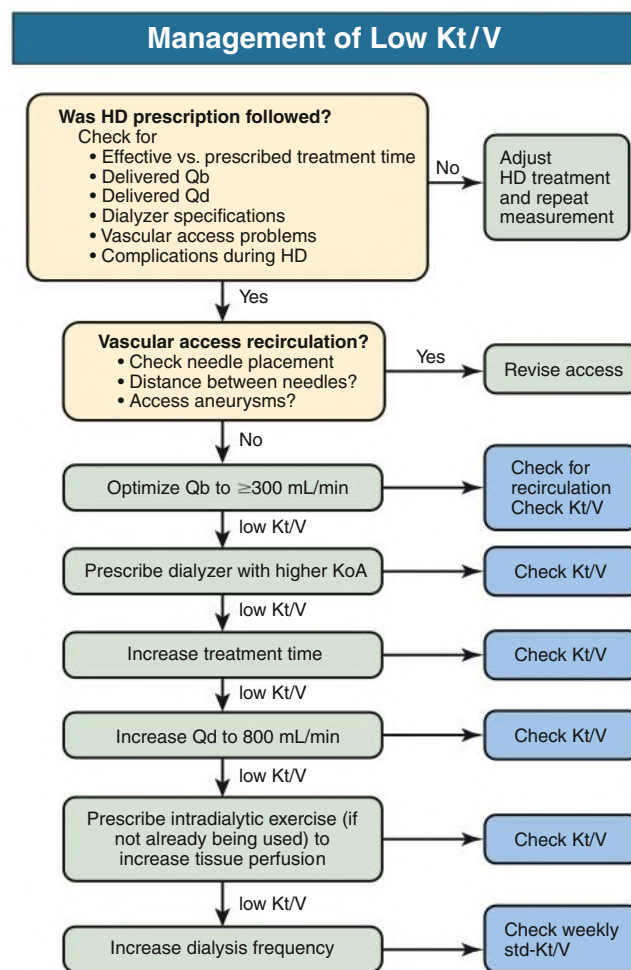


Fig. 99.4 Management of low and inadequate treatment index (Kt/V). HD, Hemodialysis; KoA , the dialyzer urea mass transfer area coefficient; Q_b , blood flow rate; Q_d , dialysate flow rate; $stdKt/V$, equilibrated Kt/V .

function with GFR greater than 3 mL/min.¹⁰ Conventional thrice-weekly dialysis remains the standard of care in most countries, where treatment time is typically governed by dialysis dose (Kt/V), with the consequence of longer dialysis in patients with higher urea distribution volumes. Acknowledging the difficulties in removing adequate amounts of fluid and low/middle molecular uremic solutes during short dialysis sessions, a minimum treatment time of 4 hours per HD session is recommended in many countries. More frequent in-center dialysis and longer dialysis hours delivered by in-center or home HD are increasingly used alternatives to conventional thrice-weekly treatment. Both modalities offer the opportunity for improved solute clearance and complete removal of interdialytic weight gain (IDWG) with fewer intradialytic problems, particularly hypotension.

Several epidemiologic studies in larger patient populations suggested a survival advantage for patients treated with extended time and/or frequency dialysis schedules, such as daily in-center or nocturnal HD 4 to 6 times a week. The randomized controlled Frequent Hemodialysis Network (FHN) study was designed to compare short daily (6 times weekly) HD vs. thrice-weekly in-center HD and, separately, nocturnal home HD 5 to 6 times weekly vs. conventional thrice weekly home HD. After 12 months of follow-up, daily in-center HD resulted in lower phosphate levels, a reduction in left ventricular mass, and an improvement in self-reported physical health.¹⁶ Analysis of patient survival over a median of 3.6 years after randomization suggested that short daily in-center HD reduced long-term mortality.¹⁷

No such benefit was observed in the nocturnal FHN trial, but this may have been due to inadequate statistical power and unexpectedly low patient recruitment.^{18,19}

Dialysate Composition

During a standard HD treatment session, a patient's blood is exposed to 120 to 200 L of dialysate. The several ingredients of dialysate composition should therefore be prescribed with great care. Generation of near-sterile dialysate and prescription of dialysate temperature are addressed in [Chapter 98](#).

Sodium

A positive sodium balance is a typical feature of end-stage kidney disease (ESKD) and an important factor in the pathogenesis of hypertension and fluid overload in HD patients. Chronic sodium accumulation may be associated with increased nonosmotic tissue sodium storage and left ventricular hypertrophy.²⁰⁻²² Judicious control of interdialytic sodium intake to approximately 5 to 6 g salt (2300 mg or 100 mmol sodium) per day is effective in normalizing blood pressure. Sodium movement during dialysis depends on the transmembrane sodium gradient between dialysate and plasma water. To avoid a positive intradialytic sodium balance, dialysate and plasma sodium concentrations should be aligned or care should be taken that dialysate sodium does not exceed the patient's average predialysis serum sodium concentration by more than 2 to 3 mmol/L. A positive intradialytic sodium balance may lead to increased interdialytic fluid intake and higher ultrafiltration volumes. However, the associations between serum and dialysate sodium levels and morbidity and mortality require further studies.

Potassium

Potassium removal during dialysis ideally should be equal to the amount accumulated during the interdialytic period. The dialysate potassium concentration must be set at a level that avoids pre-HD hyperkalemia, as well as intradialytic hypokalemia, which may provoke dialysis-induced arrhythmia. Postdialysis hypokalemia less than 3.0 mmol/L is associated with increased risk of mortality, especially in patients presenting with predialysis potassium levels less than 4.0 mmol/L.²³ The typical dialysate potassium concentration is set between 2.0 and 4.0 mmol/L. Prescription of potassium dialysate concentrations less than 2.0 mmol/L is associated with an increased risk for tachyarrhythmia and sudden cardiac death²⁴ and should be avoided.

Calcium

In light of accelerated vascular calcification in ESKD, intradialytic calcium delivery to the patient should be minimized, and in patients using calcium salts as phosphate binders, a negative intradialytic calcium mass balance is desirable.²⁵ A dialysate calcium concentration of 1.25 to 1.50 mmol/L is recommended by KDIGO, but many patients will be in positive balance even at 1.25 mmol/L. A positive calcium balance may be associated with tissue calcium accumulation, whereas lower dialysate calcium will stimulate parathyroid hormone (PTH) secretion. In most patients a dialysate calcium concentration of 1.25 mmol/L is now used.

Bicarbonate

Chronic metabolic acidosis is associated with decreased protein synthesis and increased protein catabolism and contributes to mineral and bone disorders. Pre-HD bicarbonate levels in the range of 20 to 23 mmol/L are associated with improved survival. Dialysate bicarbonate concentration is typically set between 35 and 40 mmol/L to generate a transmembrane concentration gradient favoring bicarbonate delivery

to the patient. Data from the Dialysis Outcomes and Practice Patterns Study (DOPPS) indicate that both high (>27 mmol/L) and low (<17 mmol/L) predialysis serum bicarbonate levels are associated with increased risk for mortality and hospitalization.²⁶

Magnesium

In the past, dialysate magnesium levels of 1.5 mmol/L were used frequently. Currently recommended levels are between 0.5 and 1.0 mmol/L. Whereas observational studies suggested an inverse association between serum magnesium levels and mortality,²⁷ other studies showed that magnesium dialysate does not improve intradialytic hemodynamics or abrogate myocardial stunning.²⁸ It is unclear to what extent dialysate magnesium levels have a sustained effect on serum levels.

Ultrafiltration Rate

Fluid overload is an established risk factor for the development of left ventricular hypertrophy and cardiovascular mortality. Similarly, predialysis extracellular fluid depletion is also associated with increased mortality.²⁹ Adequate HD aims to normalize body water homeostasis and extracellular volume to a postdialysis level comparable with that of patients of similar age and normal kidney function. This ideal, near-normal postdialysis body weight is frequently termed *dry weight*, which often differs substantially from the target weight prescribed. Because of the difficulties in detecting fluid overload of 2 to 3 L clinically, technical methods, such as bioimpedance, relative blood volume monitoring, and imaging (ultrasound, chest radiograph) for assessing volume status should be applied regularly.³⁰ A successful clinical approach to define dry weight has been the gradual and stepwise reduction in postdialysis weight by 0.2 to 0.5 kg per session until some cramps occur and blood pressure is in the normal range. Postdialysis target weight is then set at 0.5 kg above cramping level. UFRs above 10 mL/kg/h are associated with higher odds of intradialytic hypotension and a greater risk for mortality.³¹ It is important to realize that IDWG and fluid overload are not synonymous, and that postdialysis fluid depletion may result in high IDWG.

DIALYSIS ADEQUACY AND GOAL-DIRECTED DIALYSIS

Dialysis adequacy used to be defined as prescribing and delivering a dialysis dose that is associated with the best long-term outcome, but it has meanwhile become evident that the outcome and quality of life of HD patients cannot be influenced by dialysis prescription alone. Therefore, multiple measures and goals should be considered when prescribing dialysis, including small solute clearance, residual kidney function, volume status, biochemical measures, nutritional status, cardiovascular function, intra- and interdialytic symptoms and complications and the patient's individual experiences and goals. Patients may interpret adequacy different than clinicians, and therefore treatment goals should be individualized and reassessed over time while recognizing accepted minimums for small solute removal during dialysis.¹

The term *goal-directed dialysis* specifically refers to using shared decision-making between patient, their family/delegate, and the care team to establish realistic care goals to allow patients to meet their own life goals and allow the clinician to provide individualized, high-quality dialytic care. The aim of goal-directed dialysis is to provide the best health outcome possible for an individual on dialysis (HD or peritoneal dialysis [PD]) in terms of maintaining their clinical well-being, quality of life, and ability to meet life goals and also minimize patient and caregiver treatment burden. The components of goal-directed dialysis include those directly affected by the dialysis prescription and

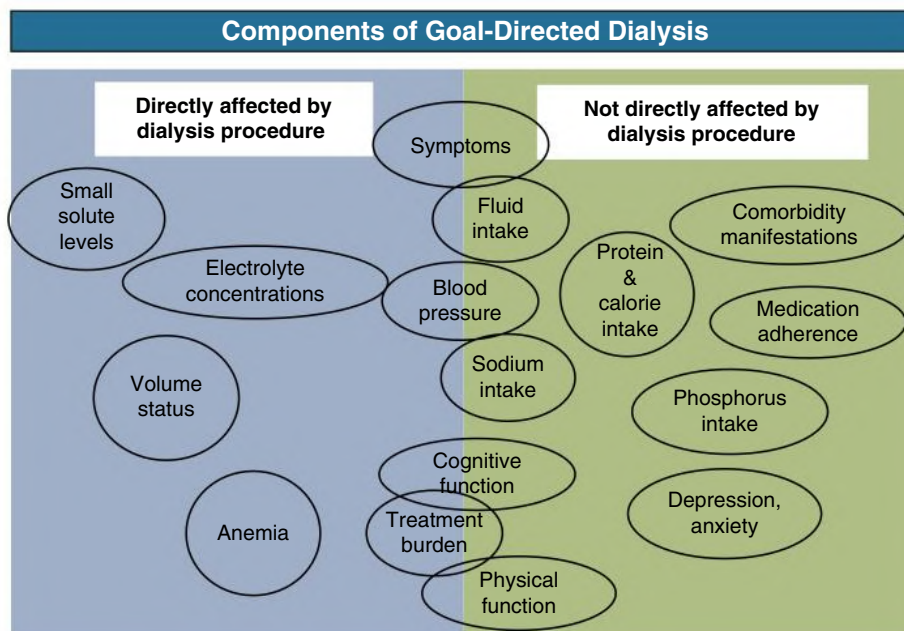


Fig. 99.5 Components of Goal-Directed Dialysis. Some individual goals can be directly affected by dialysis procedure (*left*), whereas others cannot be directly affected by dialysis procedure (*right*). Most goals, however, are overlapping the zones and can only partly be affected by dialysis procedure.

procedure, such as solute clearance electrolyte concentrations, volume status, and intradialytic symptoms, as well as those only indirectly affected by the dialysis procedure, such as symptom burden, nutritional status, activity level, work capacity, and social engagement (Fig. 99.5).³²

Prevention of Intradialytic Hypotension

Depending on the definition, intradialytic hypotension (IDH) occurs in 10% to 30% of treatments and is associated with damage to the brain, gut, heart, kidney, and poor outcome.³³ Early IDH occurring during the first half of a dialysis session is associated with particularly poor outcome and may be triggered independent of intravascular volume changes.³⁴ In contrast, IDH occurring later during dialysis is often due to high rates of ultrafiltration occasioned by the necessity to remove large IDWGs during a relatively short treatment.³¹ In addition, an individual inability to preserve organ perfusion in the face of dialysis-related hypotension or intravascular hypovolemia (e.g., secondary to uremic and/or diabetic autonomic dysfunction) may lead to short-term changes (e.g., intradialytic cardiac stunning) and long-term consequences (e.g., heart failure and cardiac mortality outcomes).³⁵ Prevention of high IDWG may help: potential strategies include dietary counseling to reduce sodium intake, lengthening of dialysis duration, dialysate cooling, use of α -sympathetic agonists such as midodrine, and avoidance of food during dialysis. In addition, prevention and management of postural hypotension immediately after dialysis is essential to prevent falls. Intradialytic hypotension is also discussed in Chapter 100.

Preservation of Residual Kidney Function

Most patients starting dialysis still have considerable RKF, but by the end of the first year, the majority may have lost RKF completely. Only 10% to 20% of patients still have RKF after more than 3 years of dialysis. A kidney clearance of 2 to 3 mL/min, which for a patient with total body water of 40 L is equivalent to a stdKt/V of 0.5 to 0.75/wk, contributes significantly to the elimination of uremic toxins with lower β_2 -microglobulin, phosphate, potassium, urea, creatinine, and uric acid

levels, as well as maintained endocrine, metabolic, and antioxidative stress functions; higher hemoglobin concentration; enhanced nutritional status; better quality of life scores; and a reduced need for dietary and fluid restrictions.³⁶

Loss of RKF is associated with left ventricular hypertrophy. Risk factors for the loss of RKF include activation of the immune system by cellulose-based membranes and dialysate water impurities; intradialytic hypotension and kidney hypoperfusion; use of angiotensin-converting enzyme inhibitors or nephrotoxic agents (e.g., radiocontrast media, aminoglycosides, and nonsteroidal antiinflammatory drugs); and hypercalcemia. Loss of RKF may be delayed by adequate target post-HD weight prescription and prevention of intradialytic hypotensive episodes.³⁷

Maintenance or Improvement of Nutritional Status

HD patients are at risk for malnutrition secondary to protein-energy wasting, decreased appetite, infection, intercurrent illnesses, hospital admissions, and missed meals after dialysis.³⁸ The recommended daily intake of 1.2 g protein per kilogram of ideal body weight and 30 to 35 kcal per kilogram of ideal body weight are often not met. The nutritional status of dialysis patients should be assessed regularly by clinical and biochemical means and by technical means such as bioimpedance. Eating during dialysis or intradialytic oral nutritional supplements including supplemental vitamins may be beneficial in patients with low spontaneous energy and protein intake. Nutrition in HD patients is also discussed in Chapter 90.

Goal-Directed Dialysis in the Elderly

Patients older than 75 years represent the largest incident and fastest-growing prevalent patient population in HD worldwide, and the unique medical and social needs of these patients require consideration. Studies in this population show that the parameters that correlate with clinical outcomes are not the traditional markers of HD dose, such as Kt/V and treatment time, but rather cardiovascular comorbidity, nutritional status, functional capacity, and frailty.³⁹ Therefore, the focus of dialysis adequacy in elderly patients should shift

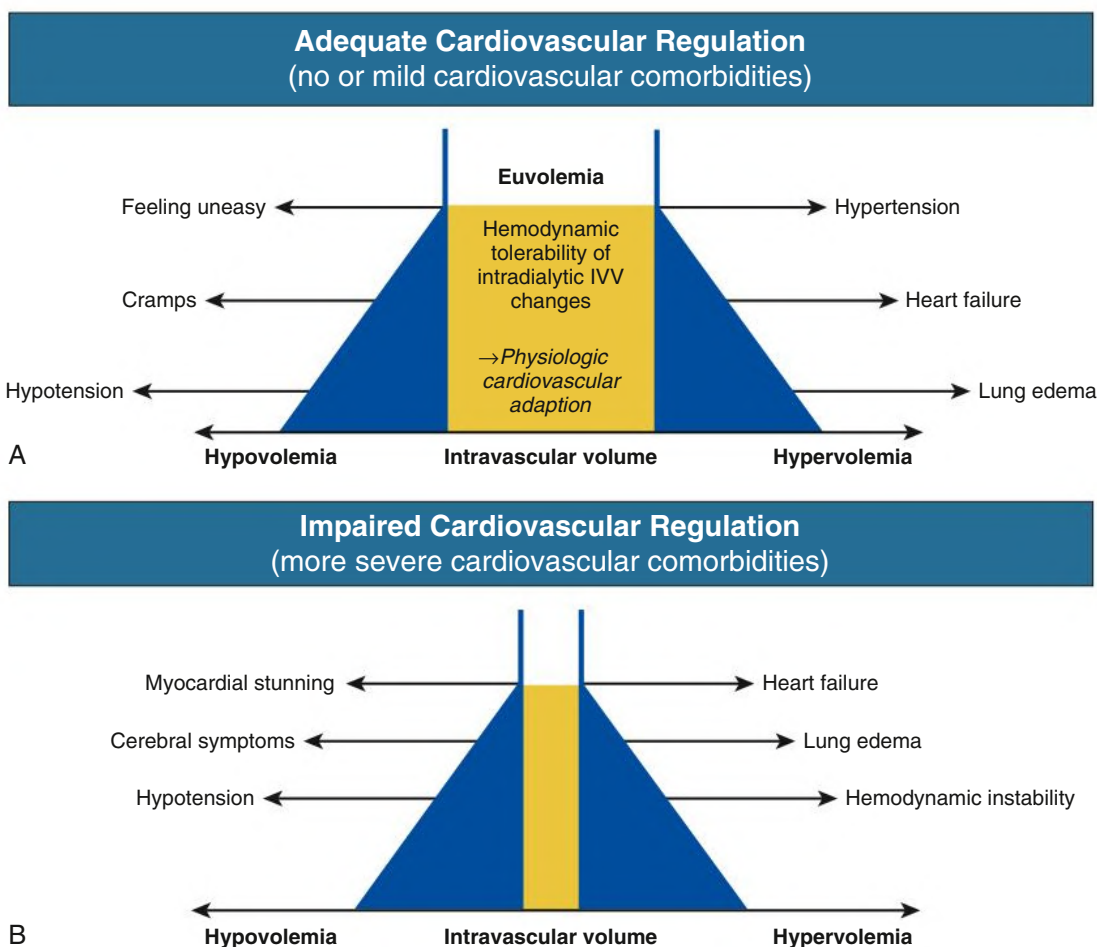


Fig. 99.6 Intradialytic Vulnerability. The occurrence of complications (arrows) in response to intradialytic changes in intravascular volume (IVV) depends on the functionality of physiologic cardiovascular (CV) adaptation mechanisms, such as an increase in heart rate and constriction of arterioles and venous capacity vessels. The individual hemodynamic tolerability of intradialytic IVV changes (yellow field) is higher in patients on hemodialysis (HD) with no or mild CV comorbidities and intact CV responses (A) than in typically older and multimorbid HD patients with impaired cardiovascular regulation (e.g., secondary to diabetic autonomic neuropathy; B). (Modified from Chauveau P, Combe C, Lavoie M, et al. Factors influencing survival in hemodialysis patients aged older than 75 years; 2.5-year outcome study. *Am J Kidney Dis.* 2001;37:997–1003.)

from urea-based assessment of dialysis dose to the use of a more multi-dimensional measure.³² Elderly dialysis patients with multiple comorbidities tolerate HD treatments less well than younger patients. Typical complications, such as intradialytic hypotension, cramps, atrial fibrillation, and postdialysis fatigue contribute to the rapid loss of independence occurring in this vulnerable patient population. The underlying etiology is reduced tolerance to intravascular volume reduction during dialysis, which is at least partially caused by impaired cardiovascular hemodynamic responses in combination with uremic and/or diabetic autonomic neuropathy (Fig. 99.6). Significant reductions in organ perfusion occurring in the gut, liver, kidneys, heart, and brain contribute to the development of inflammation, malnutrition, heart failure, loss of RKF, frailty, depression, and dementia.³³ To protect vulnerable elderly patients from the dramatic short- and long-term consequences of intradialytic tissue hypoxia, the HD prescription should be individualized. The following list provides some interventions that may help make HD treatments more tolerable in elderly patients, with the goal to reduce cardiovascular complications, sustain nutritional status, and slow the development of frailty.

- Carefully evaluate individual susceptibility to intradialytic hypotension in relation to UFR.

- Assess vascular refilling capacity by monitoring intradialytic changes in blood volume.
- Individualize maximum UFR (ideally <10 mL/kg body weight/h).
- Consider ultrafiltration profiling and/or the use of dialysis machines with biofeedback control of UFR.
- Lower dialysate temperature to prevent the intradialytic increase in body core temperature associated with an increased risk for intradialytic hypotension.
- Reduce IDWG (dietary counseling regarding salt intake, prescribe adequately dosed diuretics).
- Avoid intradialytic positive sodium balance (individualize dialysate composition).
- Increase dialysis frequency in case the patient does not tolerate a long interdialytic interval.
- Reduce β -blocker medication to allow cardiac compensation for changes in intravascular volume.
- Reduce or pause antihypertensive medication before dialysis in hypotension-prone patients.
- Assess the effect of food intake during dialysis on hemodynamic stability.
- Regularly adjust postdialysis target weight (bioimpedance).

- If hypertensive, gradually reduce postdialysis weight, avoiding intradialytic hypotensive episodes.
- Consider supine position during dialysis.
- Consider online HDF.
- Consider peritoneal dialysis in patients with otherwise intractable IDH.
- Treat intradialytic hypoxia by intradialytic oxygen application.
- Ensure three meals per day, including the dialysis day.
- Prescribe oral nutritional supplements during and between dialysis treatments.
- Prevent immobilization by early rehabilitation and physical therapy.

SELF-ASSESSMENT QUESTIONS

1. Which of the following factors related to the hemodialysis prescription in a patient without much residual kidney function is likely to result in the *worst* outcome?
 - A. Kt/V of 1.15 over months
 - B. Use of low-flux dialyzers
 - C. 3-hour dialysis times
 - D. Dialysate potassium of 2 mEq/L
2. Hypertension is very common in dialysis patients. Which factor alone is *least* likely to reduce BP?
 - A. Use of antihypertensive drugs
 - B. Longer dialysis times with fluid removal
 - C. Postdialysis weight reduction
 - D. Dietary sodium restriction
3. Which action is *not* useful when the BP falls frequently during dialysis?
 - A. Decrease dialysate temperature
 - B. Apply abdominal pressure
 - C. Reduce the ultrafiltration rate
 - D. Increase postdialysis target weight
4. The recommended target convective volume in high-volume online hemodiafiltration is 23 L/1.73 m²/treatment. Which of the following actions is *not* useful to increase the convective volume?
 - A. Increase ultrafiltration rate
 - B. Increase filtration fraction
 - C. Increase blood flow rate
 - D. Increase dialysate flow rate
 - E. Increase treatment time
5. In patients with residual kidney function, an intrinsic urea clearance of 3 mL/min is approximately equivalent to a weekly standard KtV of:
 - A. 0.10.
 - B. 0.25.
 - C. 0.75.
 - D. 1.05.
 - E. 1.50.

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